



TMDB Movie Genre Classification

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Spring 2025**

STEP 01

Data Scrapping

Data Scraping

We used [The Movie Database \(TMDB\)](https://www.themoviedatabase.com/) for scraping data.



The Dark Knight (2008)

12A | 07/24/2008 (GB) • Drama, Action, Crime, Thriller • 2h 32m

85%

User Score

👍👎

What's your Vibe?

III

♥

📺

▶ Play Trailer

Welcome to a world without rules.

Overview

Batman raises the stakes in his war on crime. With the help of Lt. Jim Gordon and District Attorney Harvey Dent, Batman sets out to dismantle the remaining criminal organizations that plague the streets. The partnership proves to be effective, but they soon find themselves prey to a reign of chaos unleashed by a rising criminal mastermind known to the terrified citizens of Gotham as the Joker.

Christopher Nolan
Director, Screenplay, Story

Bob Kane
Characters

Jonathan Nolan
Screenplay

David S. Goyer
Story

Now Streaming

Watch Now

Top Billed Cast

Christian Bale
Bruce Wayne

Heath Ledger
Joker

Aaron Eckhart
Harvey Dent

Michael Caine
Alfred

Maggie Gyllenhaal
Rachel

Gary Oldman
Gordon

Morgan Freeman
Lucius Fox

Watch Now

Silo on Apple TV+

Full Cast & Crew

Social

Reviews 15

Discussions 22

A review by maxim
★ 100% Written by maxim on December 13, 2024

I'm deeply in romantical love with the joker. this movie made my delusions worse

Read All Reviews

Media

Most Popular

Videos 9

Backdrops 111

Posters 217

Keywords

crime fighter

secret identity

anti hero

scarecrow

sadism

chaos

vigilante

superhero

based on comic

tragic hero

organized crime

anti villain

criminal mastermind

district attorney

super power

super villain

neo-noir

bold

Content Score

100

Yes! Looking good!

RangeIndex: 9994 entries, 0 to 9993

Data columns (total 18 columns):

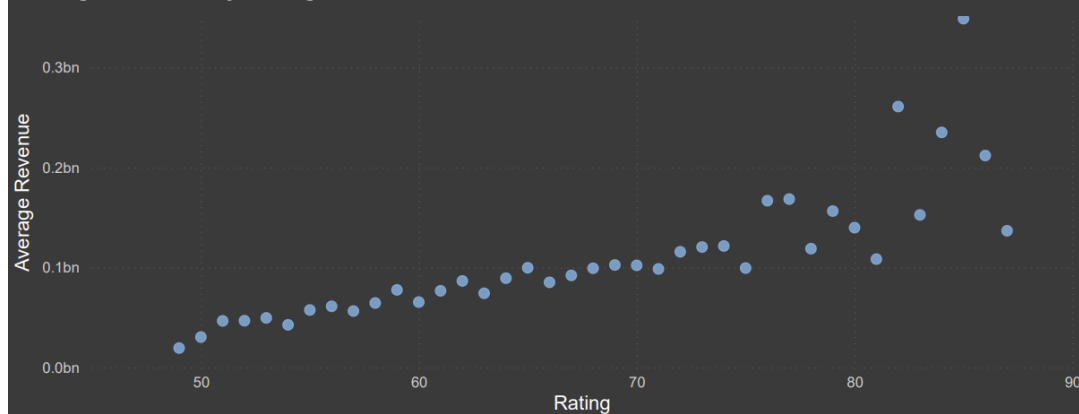
#	Column	Non-Null Count	Dtype
0	movie_name	9994 non-null	object
1	release_date	9994 non-null	object
2	rating	9986 non-null	float64
3	genre	9994 non-null	object
4	run_time	9984 non-null	object
5	certification	9904 non-null	object
6	overview	9986 non-null	object
7	tagline	8414 non-null	object
8	director	9994 non-null	object
9	language	9983 non-null	object
10	budget	9983 non-null	object
11	revenue	9983 non-null	object
12	normal_keyword_(rounded)	9994 non-null	object
13	tone_keyword_(bold)	9994 non-null	object
14	cast	9994 non-null	object
15	reviews	9994 non-null	object
16	content_score	9973 non-null	float64
17	content_score_description	9973 non-null	object

dtypes: float64(2), object(16)

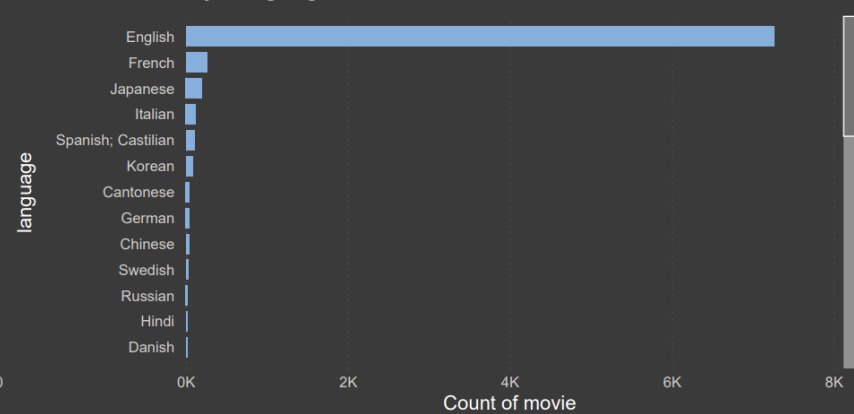
STEP 02

Explanatory Data Analysis(EDA)

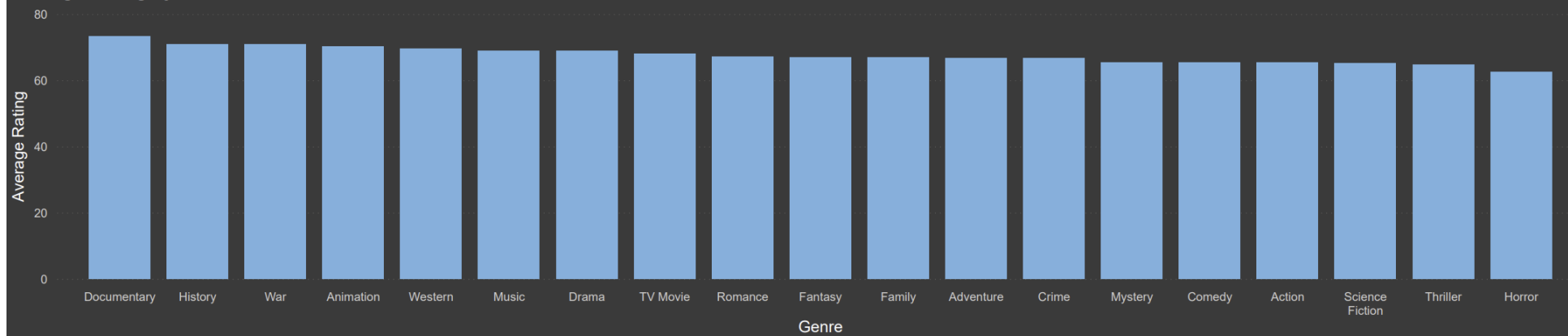
Average Revenue by Rating



Count of movie by Language



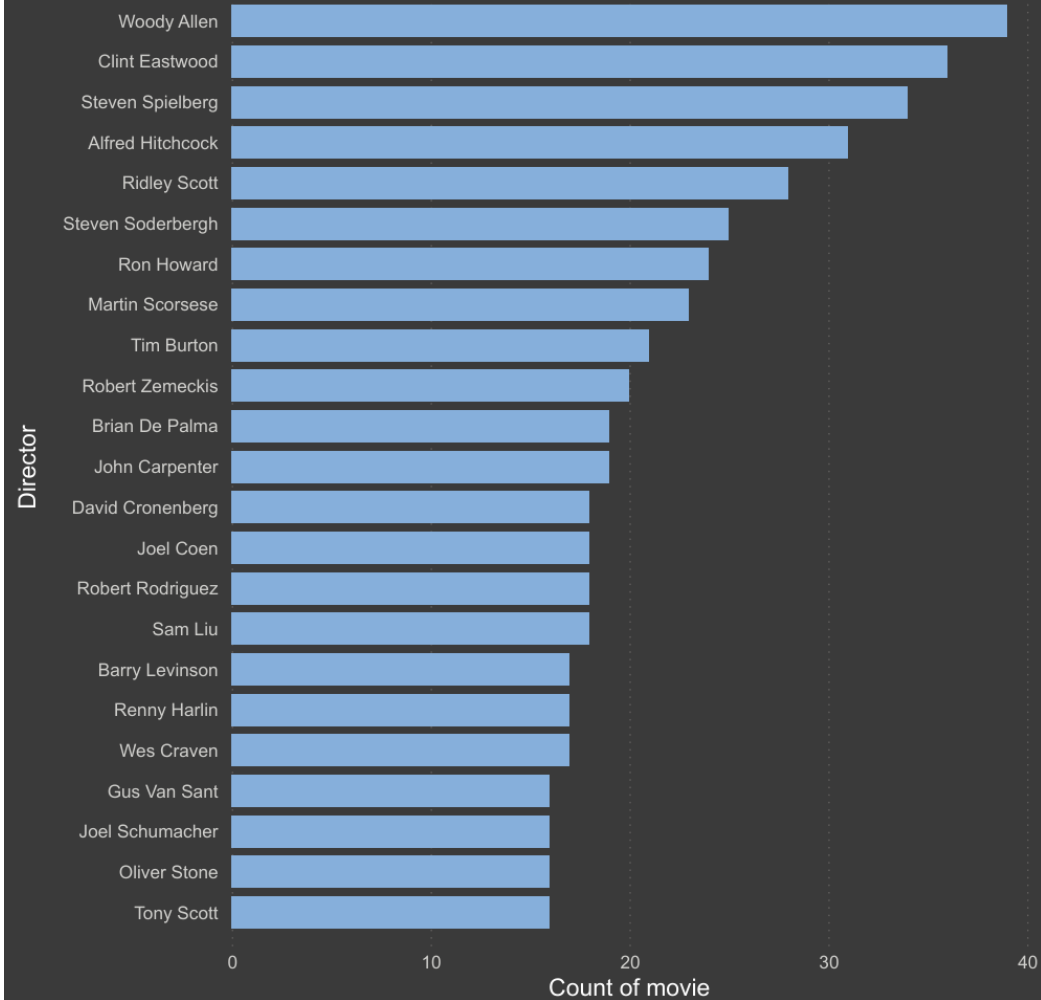
Average Rating by Genre



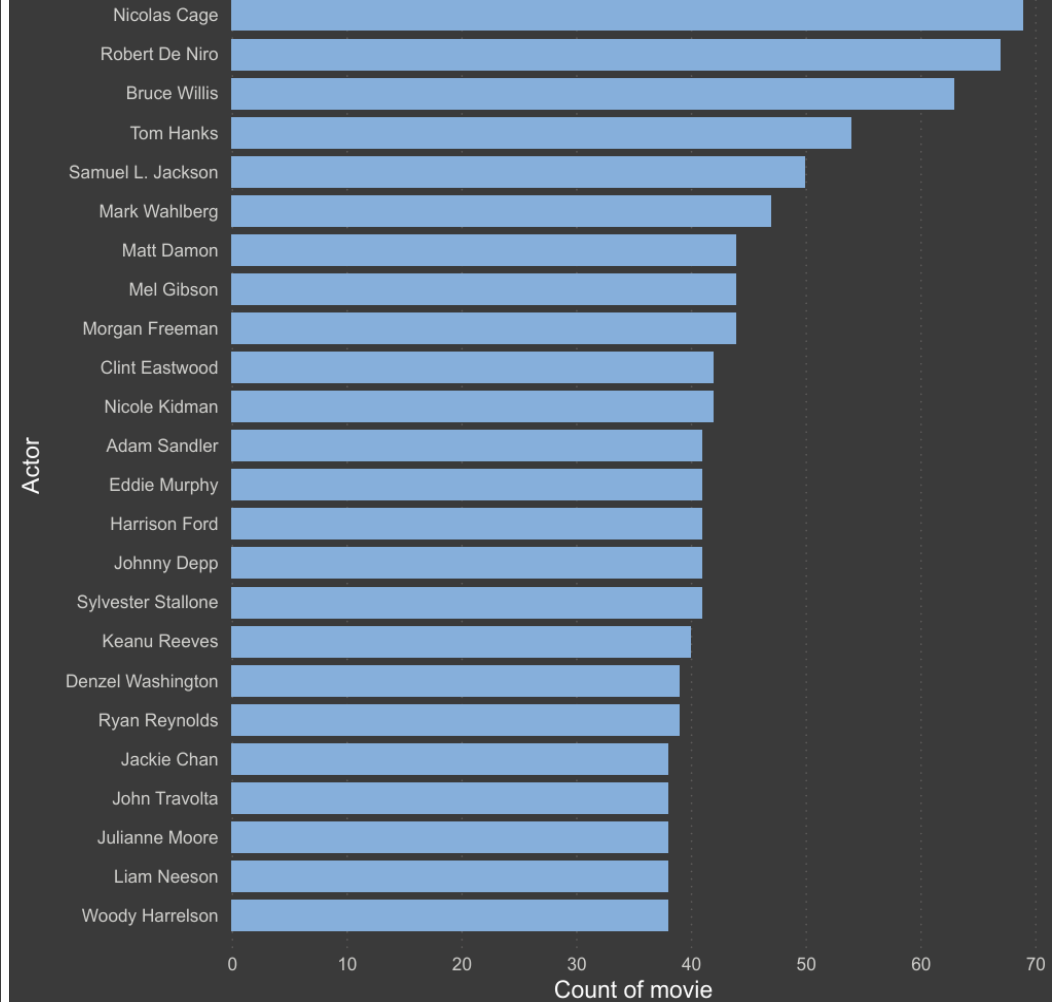
Average Monthly Revenue Trends (Since 2000)

[illegible]

Top 20 Directors by Count of Movie

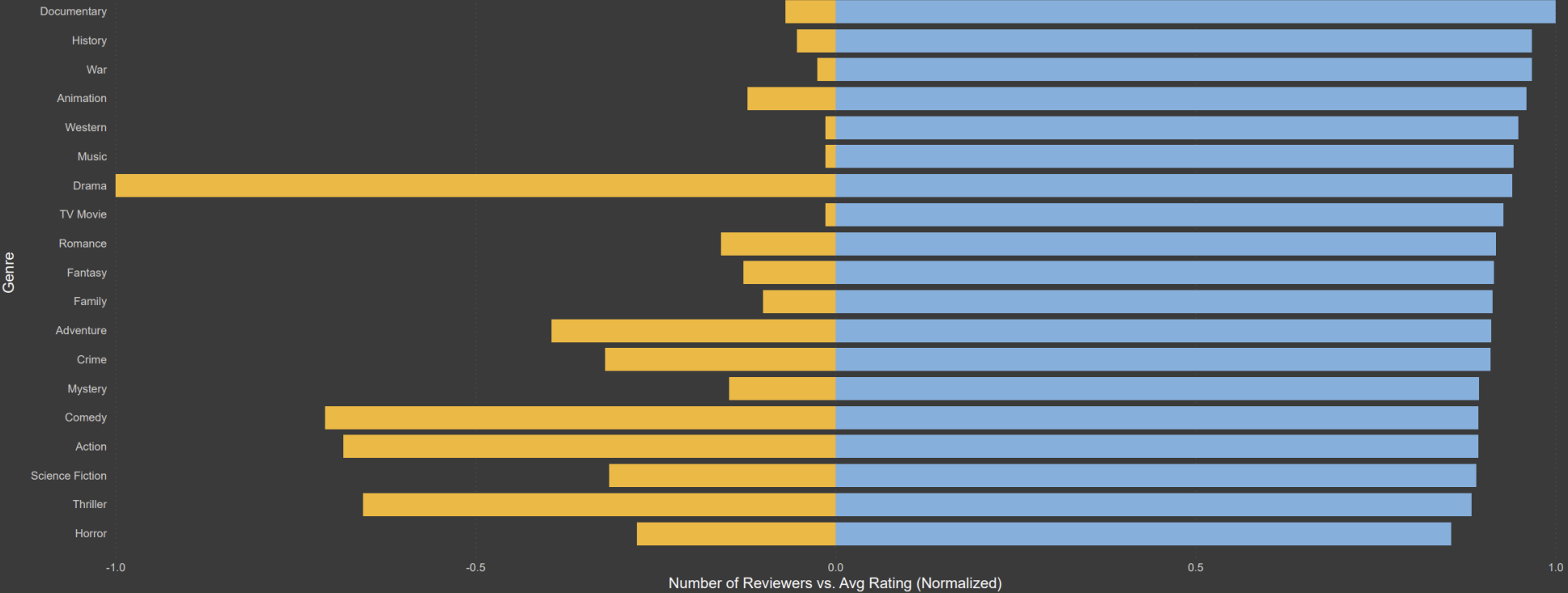


Top 20 Actors by Count of movie



Reviewers Top Genres Popularity vs. Genres Average Rating

Rating Number of Reviewers



STEP 03

Preprocessing

Steps:

1. Runtime Parsing:

- converts runtime strings into consistent numeric durations. Like '2h 30m' to total minutes.

2. Budget and Revenue Cleaning:

- Removes dollar signs and commas, handles missing values, and converts to float.

3. Release Date Correction:

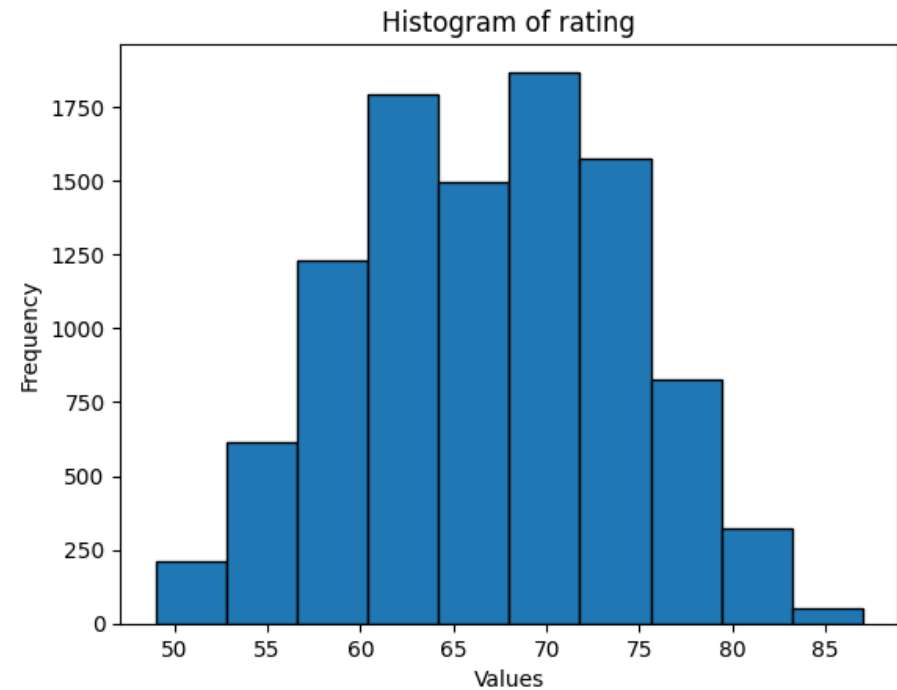
- Converts release dates to datetime objects. Like 17-Feb-95 to 1995-02-17.

4. Extracts and explodes columns with list:

- Extract columns like director, cast, genre, normal_keyword_(rounded), tone_keyword_(bold) which has multiple values for each row.

Rating column

- Since this column should be normalized or standardized, it is better to first see the data distribution and then decide which technique to use.
- Because it has a Normal Distribution, it is better to use Standardization. If it had a Uniform Distribution, we would use Normalization.
- We fill the Null values(8 values) with the mean of the column.



Certification column

- This column shows the age limit of each movie. Since the number of its unique values is large, we divided it into 7 subgroups and used ordinal encoding to label it.
- Subgroups:
 1. General Audience (Suitable for all ages)
 2. Parental Guidance Recommended (~7+)
 3. Teen Audiences (Mild to Moderate content, ~12-16)
 4. Older Teens / Restricted (~14-16)
 5. Mature / Restricted (~16-18)
 6. Adults Only (18+)
 7. Special / Regional Ratings/ Null values (Uncategorized)

STEP 04

Feature Engineering

budget_revenue_ratio column

- This is a new column whose values are obtained by dividing the budget column by revenue.
- If we take a look at the box plot of this column, we can see that it has some outlier data, we used IQR to correct this and replaced the values that were more or less than 1.5 times.
- We also fill the Null values of this column with the mean of this column.

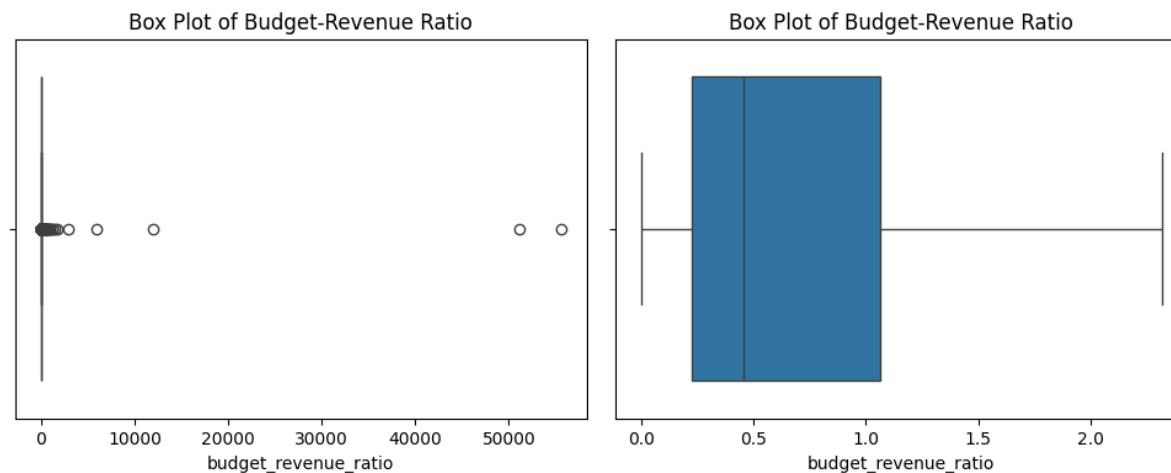
```
Q1 = final_df['budget_revenue_ratio'].quantile(0.25)
```

```
Q3 = final_df['budget_revenue_ratio'].quantile(0.75)
```

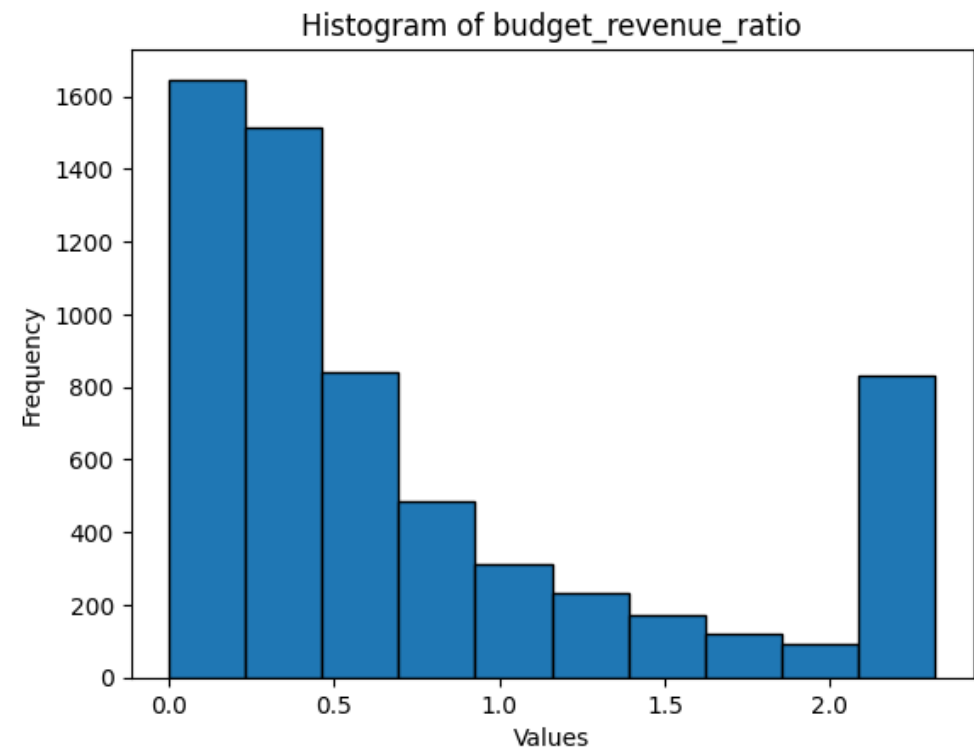
```
IQR = Q3 - Q1
```

```
lower_bound = Q1 - 1.5 * IQR
```

```
upper_bound = Q3 + 1.5 * IQR
```



- We also used standardization for this column.



Overview and Review column

- We embedded these two columns that contain text using sentence-bert, which is suitable for semantic extraction.
- Before starting the process, we filled null values with ' ' and removed extra spaces and \n
- Since each movie may have several reviews, we connected all the reviews related to one movie and then embedded them.

Actors and directors columns

- Since there are many actors and directors in the dataset, the approach we have taken to use them is to find 100 actors and 100 directors who have been repeated the most in the dataset and use them to perform one hot encoding for them.
- This means that in each movie, the actors and directors that exist have a value of 1 and otherwise a value of 0.

normal_keyword_(rounded) and tone_keyword_(bold)

- We embedded these two columns, which are the keywords of each movie, using TF-IDF

Genre column

- Since in the genre column, each movie can have several genres, we used multi labeling to encode it.
- For example, if a movie has drama and crime genres, these values will be 1 in the genre of that movie and 0 in the rest of the genres.
- Also, if the movie genre is null, it gets the label unknown.
- In total, we have 17 genres in this dataset.

Checking null value

- Since we handled null values during the preprocessing and feature engineering, as we can see, none of the columns have null values.

```
flag = True
for col in final_df.columns.to_list():
    n = final_df[col].isna().sum()
    if n>0:
        flag = False
        print(f"{col} has {n} Null values.")
if flag:
    print('No column contains null values.')
```

✓ 0.0s

No column contains null values.

rating

budget_revenue_ratio (New
Feature)

certification

overview → Text vectorization

review → Text vectorization

casts → 100 most repeated
actors

directors → 100 most repeated
directors

normal_keyword_(rounded)→
Text vectorization

tone_keyword_(bold) → Text
vectorization

round_keyword → Text
vectorization

genre (Target Column) → Multi
Labeling

STEP 05

Model Development and MLflow

MLflow Integration Overview

mlflow2.22.0

ExperimentsModelsPrompts

DNN_XGBoost_Experiments

DNN_1024-512-256_XGB_lr0.05_d8

Overview

Model metrics

System metrics

Traces

Artifacts

Description

No description

Details

Created at	06/07/2025, 08:43:19 PM
Created by	sarayetel
Experiment ID	797949132340219250
Status	Finished
Run ID	eeccf8368c7c44b8bc7dc0e4a54d55bb
Duration	2.9min
Datasets used	—
Tags	Add tags
Source	ipykernel_launcher.py
Logged models	keras +1
Registered models	Hybrid_DNN_Feature_Extractor v5 +1
Registered prompts	—

Parameters (12)	
Search parameters	
Parameter	Value
optimizer	adam
xgb_max_depth	8
learning_rate	5e-05
dnn_layers	[1024, 512, 256]
xgb_subsample	0.8
feature_model	DNN
xgb_learning_rate	0.05
epochs	80
xgb_n_estimators	200
batch_size	128
dropout_rate	0.4
classifier_model	XGBoost

Metrics (8)	
Search metrics	
Metric	Value
final_dnn_loss	0.14891809225082397
training_time	161.89224982261658
f1_micro	0.7113790504898266
recall	0.6714082503556188
final_dnn_val_loss	0.17312701046466827
score	0.7113790504898266
precision	0.7564102564102564
f1_macro	0.6048432632962026

Experiment Tracking with MLflow

Experiments  

Search experiments

- ☐ Default  
- ☒ **DNN_XGBoost_Experiments**  
- ☐ LightGBM_Experiments  
- ☐ XGBoost_Experiments  
- ☐ NN_Experiments  

DNN_XGBoost_Experiments  [Provide Feedback](#)  [Add Description](#)

Runs **Evaluation** **Experimental** **Traces**

   **Time created**  **State: Active** 

Datasets  **Sort: Created**  **Columns**  **Group by** 

	Run Name	Created	Duration	Models	Metrics
☐	 DNN_512-256-128_X...	 15 hours ago	1.2min	 Hybrid_DNN_Featu...	 0.6838597...
☐	 DNN_512-256-128_X...	 15 hours ago	1.3min	 Hybrid_DNN_Featu...	 0.6872521...
☐	 DNN_512-256-128_X...	 15 hours ago	1.2min	 Hybrid_DNN_Featu...	 0.6926406...
☐	 DNN_1024-512-256_X...	 15 hours ago	1.6min	 Hybrid_DNN_Featu...	 0.7273752...
☐	 DNN_1024-512-256_X...	 15 hours ago	2.4min	 Hybrid_DNN_Featu...	 0.7304544...
☐	 DNN_1024-512-256_X...	 15 hours ago	1.9min	 Hybrid_DNN_Featu...	 0.7309759...
☐	 DNN_512-512-512-25...	 15 hours ago	1.4min	 Hybrid_DNN_Featu...	 0.7148054...
☐	 DNN_512-512-512-25...	 15 hours ago	1.7min	 Hybrid_DNN_Featu...	 0.7183137...
☐	 DNN_512-512-512-25...	 15 hours ago	1.4min	 Hybrid_DNN_Featu...	 0.7089800...
☐	 DNN_512-256-128_X...	 16 hours ago	1.5min	 Hybrid_DNN_Featu...	 0.6671721...
☐	 DNN_512-256-128_X...	 16 hours ago	1.6min	 Hybrid_DNN_Featu...	 0.6726412...
☐	 DNN_512-256-128_X...	 16 hours ago	1.5min	 Hybrid_DNN_Featu...	 0.6723532...
☐	 DNN_1024-512-256_X...	 16 hours ago	1.9min	 Hybrid_DNN_Featu...	 0.7039161...

19 matching runs

Parameters 

☒ Show diff only

dataset	more_than_10_exa...	more_than_10_exa...	more_than_10_exa...
xgb_learning_rate	0.1	0.05	0.2
xgb_max_depth	6	8	4
xgb_n_estimators	100	200	50
xgb_subsample	1.0	0.8	1.0

Metrics 

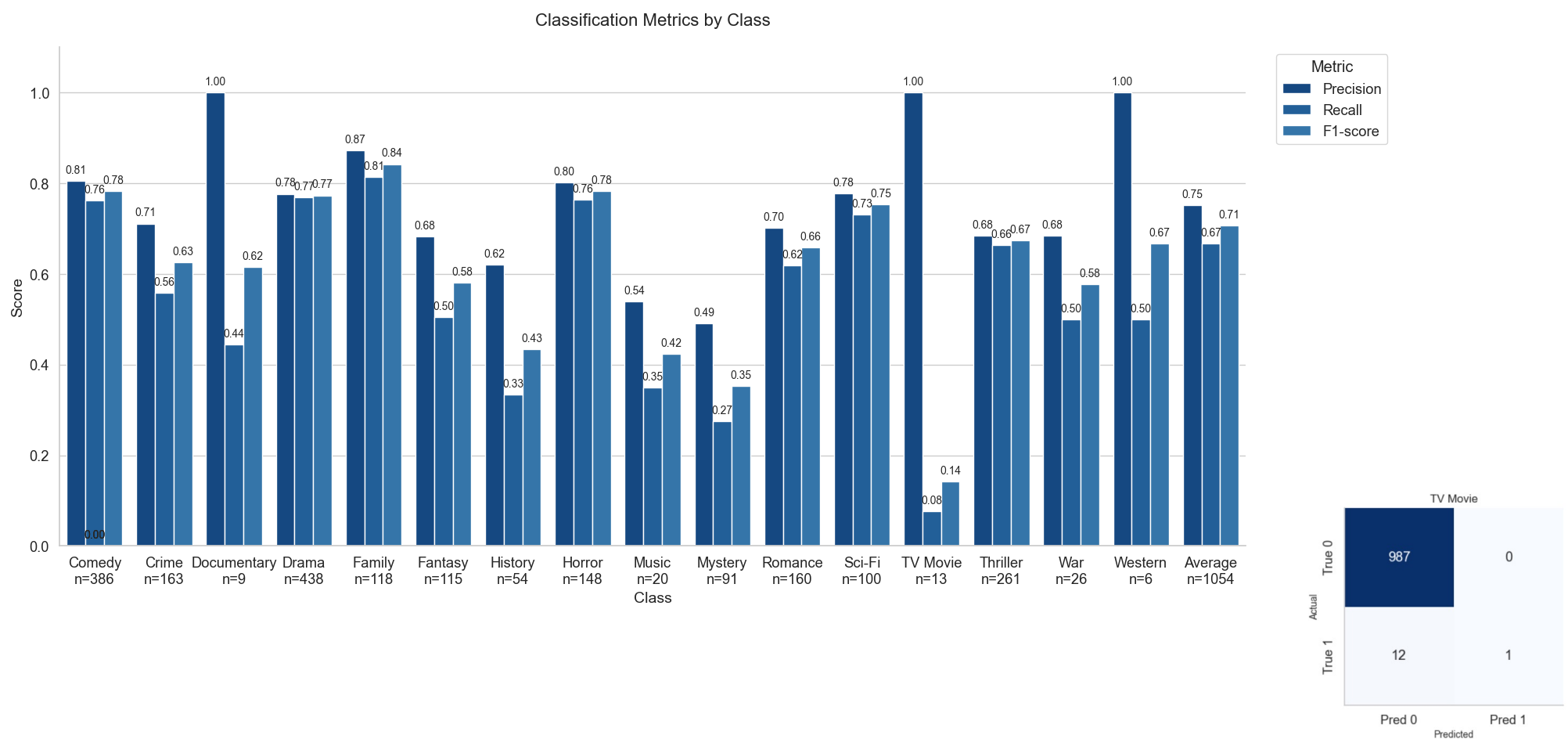
☒ Show diff only

f1_macro	0.537	0.555	0.549
f1_micro	0.731	0.73	0.727
final_dnn_loss	0.142	0.14	0.139
final_dnn_val_loss	0.162	0.16	0.16
precision	0.779	0.777	0.772
recall	0.688	0.689	0.687
score	0.731	0.73	0.727
training_time	104.7	135.4	86.47

Performance Comparison: **Neural Networks** vs. **XGBoost** vs. **LightGBM**

duration_seconds	33.74	33.09	168.5
f1_macro	0.484	0.478	0.596
f1_micro	0.691	0.718	0.677
final_train_loss	0.244	0.282	0.019
final_val_loss	0.26	0.278	0.673
precision	0.782	0.809	0.689
recall	0.618	0.645	0.666
score	0.691	0.718	0.677

Final Model : DNN + XGBoost



Unique combinations
487
124

Final Model Performance Comparison

